

SMART AND EFFECTIVE MANAGEMENT OF STREET PARKING

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COMPUTER SCIENCE



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We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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This is to certify that Project Report entitled “ Smart and Effective Management of Street Parking ” which is submitted by Anusha Goel, Pratham Saxena, Varshney Tiwari, Tarun Kumar in partial fulfillment of the requirement for the award of degree B.Tech. in Department of Computer Science of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

One of the most significant challenges in cities is parking congestion that might be caused because of the growing number of vehicles and insufficient infrastructure in the field. Drivers are usually taken long to find a parking area and this results in traffic jam, wastage of fuel and pollution. The proposed project will be called Smart and Effective Management of Street Parking, which will offer an efficient solution as the users will be able to find the available parking spaces in the area with the assistance of a web-based platform. The system enables the users to see the availability of parking spaces in real time, and choose the right parking slots, and get the best parking options according to the location and demand. The system encompasses dynamic pricing, geographical search of parking space, and administrative monitoring of parking spaces. The solution does not rely on physical sensors since they rely on manual and system-based updates on slots. This renders this system cheap and simple to execute. The project is created with the help of the latest web technologies and is supposed to make parking effective and decrease traffic jams as well as increase the comfort of users.

TABLE OF CONTENTS	Page No.
DECLARATION.....	ii
CERTIFICATE.....	iii
ACKNOWLEDGEMENTS.....	iv
ABSTRACT.....	v
LIST OF TABLES.....	viii
LIST OF FIGURES.....	ix
LIST OF ABBREVIATIONS.....	x
SDG Mapping.....	xi-xii
 CHAPTER 1 INTRODUCTION	 1-7
1.1 Introduction to Project	1
1.2 Project Category	3
1.3 Objectives	4
1.4 Structure of Report	6
 CHAPTER 2 LITERATURE REVIEW	 8-14
2.1 Literature Review	8
2.2 Research Gaps	12
2.3 Problem Formulation	13
 CHAPTER 3 PROPOSED SYSTEM	 15-22
3.1 Proposed System	15
3.2 Unique Features of The System	19
 CHAPTER 4 REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION	 23-41
4.1 Feasibility Study (Technical, Economical, Operational)	23-27
4.2 Software Requirement Specification	27
4.2.1 Data Requirement	27
4.2.2 Functional Requirement	28

4.2.3	Performance Requirement	29
4.2.4	Maintainability Requirement	30
4.2.5	Security Requirement	30
4.3	SDLC Model Used	31
4.4	System Design	32
4.4.1	Data Flow Diagrams	33
4.4.2	Use Case Diagrams	36
4.5	Database Design	38
CHAPTER 5 IMPLEMENTATION		41-45
5.1	Introduction Tools and Technologies Used.	41
5.2	Dataset Description	43
CHAPTER 6 TESTING, AND MAINTENANCE		46-53
6.1	Testing Techniques and Test Cases Used	46
CHAPTER 7 RESULTS AND DISCUSSIONS		54-61
7.1	Presentation of Results (Charts/Graphs/Tables)	54
7.2	Performance Evaluation	60
7.3	Key Findings	61
CHAPTER 8 CONCLUSION AND FUTURE SCOPE		62-64
REFERENCES		65
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Project Report Plagiarism Report		

LIST OF TABLES

Table 4. 1 Technical Feasibility Analysis	24
Table 4. 2 Economic Feasibility Analysis	25
Table 4. 3 Operational Feasibility Analysis	26
Table 4. 4 Database entities Description	39
Table 5. 1 Tools and Technologies used	42
Table 6. 1 Sample Test Cases	48
Table 6. 2 Test Environment Configuration	49
Table 6. 3 Maintenance Activities	52
Table 7. 1 Vehicle Detection Accuracy	54
Table 7. 2 Parking Occupancy Estimation	55
Table 7. 3 System Performance Metrics	59

LIST OF FIGURES

Figure 4. 1 View of Spiral Model.....	32
Figure 4. 2 Zero Level DFD.....	34
Figure 4. 3 Level One DFD.....	36
Figure 4. 4 Use Case Diagram.....	38
Figure 7. 1 Vehicle Detection Accuracy across Different Video Frames.....	54
Figure 7. 2 Parking Occupancy During Different Time Periods.....	56
Figure 7. 3 Parking lot visuals during morning hours.....	57
Figure 7. 4 Parking lot visuals during afternoon hours.....	57
Figure 7. 5 Parking lot visuals during evening hours.....	58
Figure 7. 6 Availability of Free Slots Across Test Scenarios.....	58
Figure 7. 7 Overall System Performance Evaluation.....	59

LIST OF ABBREVIATIONS

AI – Artificial Intelligence

ML – Machine Learning

CV – Computer Vision

R-CNN – Region-Based Convolutional Neural Network

Mask R-CNN – Mask Region-Based Convolutional Neural Network

YOLO – You Only Look Once

CNN – Convolutional Neural Network

DNN – Deep Neural Network

IoT – Internet of Things

API – Application Programming Interface

SDLC – Software Development Life Cycle

UX – User Experience

DB – Database

NoSQL – Non-Relational Database

FPS – Frames Per Second

SDK – Software Development Kit

GPS – Global Positioning System

IDE – Integrated Development Environment

RTDB – Realtime Database

COCO – Common Objects in Context Dataset

SDG Goal	SDG Title	Project Contribution
SDG 9	Industry, Innovation and Infrastructure	The project creates an AI-based intelligent parking manager system that operates on the computer vision, deep learning, and cloud technologies. It aids modernization of city infrastructure, as the manual and sensor-based parking system substitutes with the cost-effective and scalable digital one..
SDG 11	Sustainable Cities and Communities	The system enhances the movement of persons in the city as it will supply real time information on the availability of parking space. It will decrease traffic jams, elevate the use of street parking, efficiency in terms of transportation, and help make our cities smarter and more sustainable.
SDG 12	Responsible Consumption and Production	The technology will guarantee the best use of the available parking facilities by preventing unnecessary movement and functionality of vehicles besides better management of parking spaces. it assists in curbing the abuse of infrastructure and induces effective resource distribution..
SDG 13	Climate Action	The system assists in the reduction of fuel consumption and carbon emission by minimizing the amount of time spent by vehicles to find parking spots. This helps in saving the

		environment and also reduce air pollution in towns.
SDG 8	Decent Work and Economic Growth	The effective parking control increases the accessibility of commercial zones and infrastructure. The system contributes to productivity through minimizing the traffic delays and promotes innovations based jobs in intelligent transportation systems..

CHAPTER 1: INTRODUCTION

1.1 Introduction

The process of urbanization is among the most prominent features of the twenty first century. The swift industrialization, movement to the urban centers, and living standards have greatly augmented the number of cars on the roads. In the developing economies like India, the number of vehicles owned privately has been increasing much higher than the development of the roadways and other parking areas. This is causing severe issues of parking challenges in urban centers that directly affect the traffic management and urban mobility.

The problem of parking has become one of the biggest causes of traffic congestion in the cities. Research has shown that a significant part of the traffic in the city is a fleet of vehicles that go around in the pursuit of parking spots. Road users usually take some minutes even hours trying to find an empty space in the busy places. The result of this scenario is increased traffic jams, wastage of fuel, air pollution, increased stress among drivers and wastage of time that could otherwise be used productively. When all these factors are cumulated, their impact on environment and general efficiency of transportation systems in the city is negative.

The conventional parking management systems are mainly manual or semi-automated. In most places, parking attendants enter in and out car details manually and customers depend on their eyes to check out the availability of slots. These systems do not have real-time monitoring and cannot be dependable to give precise updates on the occupancy status. Moreover, the coordination between the various parking areas is weak, there is poor utilization of the available street parking, there is no centralized control and the level of scalability is also low. These constraints render traditional systems as insufficient in dealing with dynamic nature of present day urban settings.

The problems are even more acute when it comes to parking in the streets. Parking in the streets is often uncontrolled and unstructured and it results in unlawful parking, parking more

than once, and blocking of the roads. The lack of a monitoring system to keep a check on the occupancy in real time makes the authorities struggle with the laws regulating the parking and the smooth flow of traffic. There is also the lack of technological integration which contributes even more to the inefficiencies and restricts effective planning.

To solve the above problems, the Smart and Effective Management of Street Parking System proposes an intelligent, web-based, and AI-driven solution. It is a system intended to track and control parking spaces in real-time and use the best computer vision and cloud technologies. It has the main purpose of offering real-time and precise parking availability to the users via the online platform and, consequently, lessening the necessity of manual assistance and costly hardware setups.

The system uses a video-based vehicle detection system that is fuelled by the deep learning algorithm of Mask R-CNN. The model recognizes vehicles and can decide whether they are occupied or not by analyzing the frames taken by surveillance cameras located in parking places with high accuracy. In contrast to classical methods of detection using bounding boxes exclusively, the Mask R-CNN is an instance segmentation making it possible to identify vehicles on a pixel level. This improves accuracy in the detection especially in crowded or partially blocked surroundings. The resulting processed data is then updated to a cloud database whereby the users can see the real time parking availability using a web interface.

There are advantages to this proposed system. This method does away with physical sensors at each of the individual parking spaces and therefore reduces capital outlays for equipment and maintenance costs. Additionally, it provides centralized monitoring capabilities, real-time status updates, improved traffic flow and more effective usage of available parking space. Indirectly, the proposed method will decrease fuel consumption and reduce carbon emissions through lessened driving time spent searching for parking spaces.

In addition to that the proposed solution can be correlated to a new generation of smart cities, where various digital technologies can be integrated into urban infrastructure and provide

citizens with better access to city public services. The proposed solution uses AI, ML, CV, and other web technologies for a scalable and efficient management system for all parking needs. In addition to the solution for parking problems in the short term, the proposed platform will assist in solving the problem for the long term as well, using dynamic pricing models; demand forecasting with predictive analysis; digital payment services on the platform; and GPS navigation assistance.

To sum up, the Smart Street Parking Management System is not a software solution but an all-encompassing urban mobility solution. It is meant to enhance efficiency, sustainability, and convenience to users in the modern cities by turning the traditional parking management into an intelligent and data-driven system that can address the changing needs of the transportation in the cities.

1.2 Project Category

The Smart Street Parking Management System that is suggested belongs to the category of Smart City Applications that are based on Artificial Intelligence. It is an interdisciplinary project that combines the ideas of Artificial Intelligence, Machine Learning, Computer Vision, Cloud Computing and Web Technologies into a solution to the real life problems of urban mobility. The system is tailor-made to tackle the problem associated with parking management in cities in terms of automation and making smart decisions.

Technologically speaking, the project is part of Intelligent Transportation Systems (ITS). Intelligent Transportation Systems are aimed at increasing the efficiency, safety and sustainability of transportation through the use of sophisticated computational technologies. Through its property of machine learning, including Mask R-CNN, as a vehicle detector, the system is an excellent example of how AI is practically applied to traffic and parking management. Instance segmentation ensures that vehicles in parking spaces are properly identified, and the solution is more trusted as compared to the manual and traditional sensor-based systems.

As far as the type of application is concerned, we can classify the project as a real-time monitoring and automation system. It takes live or recorded video streams obtained by surveillance cameras, analyzes them with the help of computer vision algorithms, and changes the information on parking occupancy in real time. This is due to the fact that cloud-based databases have been integrated to make the system a distributed and synchronized platform that can be accessed with multiple devices. Such dynamic real-time capability classifies the project as a dynamic, data-driven management system.

The project can be defined as a research-based development project, in the sphere of Artificial Intelligence and Smart Infrastructure, academically. It provides the application of deep learning models in real-life setting and emphasizes how AI can substitute the hardware-based systems. The physical parking sensors removal also further classifies the project as a cost-reduced software-focused solution.

In the industry perspective, the system falls under category of Smart Mobility and Urban Infrastructure Management. It facilitates the activities of smart cities by allowing effective use of parking facilities, alleviating traffic jams, and lessening the effects on nature. The solution can be offered in large scale and can be implemented in public parking lots, business areas, educational facilities, and street parking areas.

Thus, the Smart Street Parking Management System falls into the overall category of AI-based Intelligent Transportation and Smart City solution that revolves around the real-time monitoring, automation, and cloud-integrated urban infrastructure management.

1.3 Objectives

The key goals of the project are:

- To create an effective and scalable parking management platform.
- To give real-time parking availability information.
- To minimize traffic jam due to search of parking places.

- To add the comfort to the users and their time.

The goals can be further extended in the following way:

Creation of a Parking Management Portal

The system will seek to establish centralized platform, where all the parking related data will be stored, processed and presented. This provides equal control and supervision of parking lots.

Availability Monitoring in Real Time

Among the major objectives is to give precise real-time feedback on the occupancy of parking slots through the use of computer vision in detecting vehicles.

Minimization of Traffic Congestion

The system helps decrease the overall traffic in congested regions since it reduces the unnecessary vehicle movement that is due to searching a parking space.

Cost-Effective Implementation

The system removes the reliance on hardware-based sensors and allows the system to be deployed at a very low cost together with maintenance.

Improved User Experience

The site has been engineered to have a convenient interface to enable the user to easily find the parking space, peruse the availability and also choose the right parking space.

Effective Use of Parking in the Streets

Parking spaces in the streets are not used fully because they are not visible. The system assures maximization of the use by availing clear occupancy information.

Administrative Monitoring Support

The system will have an administrative dashboard that will manage the parking slots, occupancy and updates on the slot information.

Scalability and Enhancement Capability of Future

There will be the possibility of expanding the architecture in the future via AI-based predictive models, incorporation of mobile apps, and automated billing solutions. It is through these objectives that the system will offer a quality, intelligent, and sustainable solution to the problem of parking in the city.

1.4 Structure of Report

The report is divided into several chapters each of which is dedicated to a certain stage of the project development process.

Chapter 1: Introduction

This chapter gives a summarization of the parking issue, the motivation behind the project, goals and classification of the project.

Chapter 2: Literature Review

This chapter talks about the past studies and about the systems of parking management. It outlines shortcomings of existing solutions and points out gaps in research, which will explain the necessity of the suggested system.

Chapter 3: Proposed System

This chapter explains the design, structure, and the mode of operation of the proposed parking management system. It describes system flow and peculiarities.

Chapter 4: Requirement Analysis and System Specification

This chapter gives the feasibility analysis, software requirement specification, system design diagrams and database design.

Chapter 5: Implementation

The tools and technologies employed, description of the dataset, development environment, and the combination of AI components with the web platform is explained in this chapter.

Chapter 6: Testing and Maintenance

This chapter outlines testing methods, validation, performance testing and maintenance plan.

Chapter 7: Results and Discussions

In this chapter, the results of the system performance are provided, the metrics of detection accuracy are given, the occupancy prediction data is offered, and the main discoveries of the project implementation are provided.

Chapter 8: Conclusion and Future Scope

The chapter will provide an overview of the project success, and the possible improvement and future suggestions.

CHAPTER 2: LITERATURE REVIEW

2.1 Literature Review

The One area that has received extensive research in the development of smart cities is parking management. As the density of vehicles in the cities grows at a very high rate, scientists and engineers have forwarded to develop several technological concepts to achieve better efficiency in parking, minimize congestion, and urban mobility. This has developed over the years to various methods of sensor system hardware to intelligent platforms that are AI-driven. The section entails a thorough literature review of the current parking management technologies and methodologies.

2.1.1 Sensor-Based Systems

The previous parking management systems used were largely dependent on physical sensors that were placed within the slots of the parking. Some typical examples of sensors are Ultrasonic Sensors, Infrared (IR) Sensors, Magnetic Sensors & Pressure Sensors.

The ultrasonic sensors can detect the presence of vehicles based on the distance between the sensor and the object placed over the sensor. IR sensors are used to detect motion or heat pattern, and the magnetic sensors are used to detect any interference in the magnetic field of the earth by metallic vehicles. The systems are typically precise in their vehicle presence detection and they are commonly applied in business parking lots and in multi-level parking buildings. But they have a number of limitations like expensive cost of installing one parking slot, elaborate wiring and infrastructure specifications, regular maintenance because of being exposed to environmental conditions, large outdoor areas have scalability problems, street parking is hard to install.

In open street parking, installation of sensors under roads means that there are civil changes in the roads, which are expensive and time consuming to deploy. Also, a sensor malfunction may create false occupancy data, which decreases the reliability of the system. In spite of these,

sensor-based systems formed the basis of intelligent parking technologies by bringing in automation in occupancy detection.

2.1.2 RFID-Based Parking

Parking management has also been subject to the Radio Frequency Identification (RFID) technology. Under RFID systems, RFID tags are installed in vehicles and RFID readers installed in the entrances of parking. The reader reads the tag and logs the movement when a vehicle is entering or leaving a parking facility. RFID-based systems have the following advantages, such as Automated check-in and check-out, reduced human intervention, better security and tracking of vehicles, fast identification process.

RFID systems are also efficient in a controlled setting like private parking lots, company campuses and gated communities. They are, however, limited when used in the management of open street parking: RFID is mainly used to monitor the movement of vehicles rather than slot occupancy. It has no visual confirmation as it can be used to determine whether a slot is occupied. Also, it imposes that the vehicle owners install RFID tags and organization of infrastructure may be costly. Thus, although the RFID enhances the ability to control the access and observe the vehicles, it cannot address the issues of the real time visibility of street parking on its own.

2.1.3 IoT-Based Smart Parking

As the Internet of Things (IoT) technology developed, the use of sensors in the parking system with cloud platforms started to monitor the parking remotely and have a centralized control. On smart parking systems based on IoT sensors detect occupancy, information is relayed to cloud servers, availability is available to the users through mobile or web-based applications. IoT platforms allow, real-time monitoring, centralized data storage, remote accessibility.

Data analytics capabilities are utilized by the company to improve the value of its loans. Data analytics capabilities The company uses data analytics capabilities to enhance the value of loans.

Nevertheless, systems based on IoT continue to rely extensively on hardware. Every parking slot needs a sensor node, a communication module and a power source. This brings with it a number of real life issues ,high hardware cost, battery replacement issues, dependency network connectivity.

Hardware maintenance complexity Hardware maintenance refers to the process of sustaining hardware equipment, software, and applications to ensure their correct operation within an organization (Muller et al 2008).

Environmental vulnerability: According to this theory, environmental factors such as the physical environment, water quality and wildlife influence marine ecosystem sustainability.

Using IoT sensors on each parking slot is not economical when it comes to large urban street parking. Therefore, IoT enhances accessibility of data but the hardware nature restricts its scalability in terms of street parking in cities.

2.1.4 The Computer Vision-Based Systems

The innovations of Artificial Intelligence and Computer Vision in the recent past have had a major impact on the parking management systems. The camera-based systems are used instead of physical sensors installed on each parking slot and working based on image processing and deep learning algorithms to identify the presence of vehicles.

The systems based on computer vision usually work under Surveillance cameras,video frame capture, Image preprocessing,object detection based on deep learning.A number of algorithms of deep learning have been used to detect vehicles:

A CNN (Convolutional Neural Networks) represents a type of neural network that specializes in solving linear programming problems.A CNN (Convolutional Neural Networks) is a kind of neural network that is specialized in linear programming.Its types are R-CNN (Region-Based CNN), Faster R-CNN, Mask R-CNN.

CNN methods examine the visual patterns and categorise objects. R-CNN proposed the use of region proposals to be more precise in object localization. R-CNN was enhanced by faster R-CNN. YOLO minimized the real-time detection without having to repeat classification and localization steps.

Mask R-CNN also improved the level of detection by adding instance segmentation, which enabled the object within the image to be recognized on a pixel-by-pixel basis. This allows the accurate separation of overlapping cars in busy parking spaces. The benefits of Computer Vision-Based Systems include -slot-level sensors are not necessary, reduced cost of hardware (cameras were all that was needed), expansive in terms of size, fully automated and able to identify various types of vehicles, centralized monitoring. These systems however need: High computational power, proper lighting conditions, real-time performance optimized algorithms.

Nevertheless, computer vision solutions are nowadays regarded as the most promising and scalable solution to smart street parking management.

2.1.5 Reservation Systems web-based.

Another parking product line is the web-based reservation systems. These websites enable customers to reserve parking spaces. These systems are frequently found in malls, airports and personal parking lots. Major characteristics of online booking systems are: Online booking, Slot reservation, payment integration, user account management.

In spite of the fact that these systems make it more convenient, there are some limitations. They also use manual updates of slots, they do not have real time occupancy detection, they do not necessarily check whether a slot is occupied or not.

Therefore, no reservation system can provide the correct parking availability in the absence of integration with real-time detection mechanisms.

2.1.6 Deep Learning Approaches

Deep learning has become an effective technology in smart parking. Instances of such algorithms include Mask R-CNN, which allow instance segmentation, where every object detected is not only a bounding box at pixel scale. Mask R-CNN provides proper object identification.

Deep learning models that are trained on datasets like COCO (Common Objects in Context) are able to identify several classes of vehicles including cars, motorcycles, buses, trucks. This generality thus renders the deep learning to be very apt to street parking scenarios. Upon reviewing the available literature.

2.2 Research Gaps

There exist a number of research gaps in the sphere of smart street parking management despite the new technologies which have been introduced in terms of parking.

The current smart parking technologies have a number of limitations which diminish their efficiency in large cities. Parking solutions that are sensor-based are expensive to install and maintain since most systems that are based on hardware need to install its own sensors on each parking slot, which adds major costs to infrastructure. The systems also lack scalability particularly when they are applied on a large city area because of the size, complexity, and the maintenance of the IoT devices. As well several intelligent parking systems are provided to structured parking centers like shopping centers and parking lots, whereas open street parking is insufficiently covered. The traditional and semi-automated systems become unable to deliver sound real-time occupancy information, which results in faulty parking availability information. The second significant disadvantage is the absence of a smooth connection between artificial intelligence and cloud technologies, which limits the possibility of processing data in real-time and monitoring it centrally. In addition, some of the current systems place a lot of emphasis on the mechanisms of detection at the back-end and do not emphasize on user experience leading to the designing of systems that are poorly designed and less friendly to user-interfaces. Most of the solutions do not also have centralized administration dash boards to monitor and control multiple parking areas, which lead to

inefficiencies of dealing with dynamically urban parking situations. Furthermore, there are still real-life environmental conditions like lighting effect, shadows, and vehicle blockages that influence the accuracy of the detection, which make the vision-based parking detection systems less reliable. These gaps indicate that a holistic, integrated, and scalable system that integrates intelligent detection with cloud access and user-friendly design should be created.

2.3 Problem Formulation

The growing pressures in the urban areas in terms of parking space requirements have established massive discrepancies in the infrastructure capacity and the density of vehicles. In essence, the fundamental issue can be outlined as to design and deploy an economical, scalable, and real time smart street parking management application that precisely identifies vehicle parked status and enables vehicle users to receive real time parking availability data.

The architecture of the suggested system is a Web-Based Smart Street Parking Management System based on Artificial Intelligence and Cloud Computing and includes: Camera Input Layer, AI Detection Layer (Mask R-CNN), Processing Layer (python + OpenCV), Cloud Database Layer (Firebase RTDB), Web Interface Layer (React.js), Admin Dashboard.

Working Process: Frame is recorded with the help of surveillance cameras, it is processed with the help of OpenCV and is resized and pre-processed, vehicles are detected with the help of mask R-CNN by means of COCO dataset labels, quantity of detected vehicles is counted, occupancy and free slot is estimated.

Firebase Realtime Database is updated with data, and users can see live availability by using the web application, and an administrator can locate parking status using the dashboard, the system would remove the need to rely on physical slot sensors, provide a proper real-time occupancy estimate, decrease traffic jams, save time using this system in the city, and ensure the system is not expensive to deploy.

Therefore, the formulation of the problem is concentrated on the incorporation of Artificial Intelligence, Cloud Computing, and Web Technologies into a single parking management system that would be able to cope with the challenges of the contemporary urban environment.

CHAPTER 3: PROPOSED SYSTEM

3.1 Proposed System

The system suggested is a Web-Based Smart Street Parking Management System with Artificial Intelligence and Cloud Computing. The system will be used to give real time information about the availability of parking spaces through computer vision methods as opposed to the traditional hardware sensors. The overall idea behind the proposed system is to develop an efficient, scalable, and affordable system of managing street parking within urban environments.

The system will combine various technologies such as deep learning, cloud databases and the latest web development frameworks to make it run smoothly. With the help of vehicle detection based on video, the system removes the use of physical slot sensors and allows monitoring parking spaces centrally.

The system architecture is broken down into a number of logical layers to guarantee modularity, scalability as well as maintainability. The layers have a certain functionality and communicate with each other via clearly defined interfaces.

All the layers are elaborated below.

3.1.1 Camera Input Layer

Camera input layer will capture real time video stream of the surveillance cameras that will be installed in the parking lots. These cameras are placed in strategic positions and they encompass as many parking slots as they can in their field of view.

The important features of this layer are continuous video streaming, frame capture of high resolution, broad-based coverage of parking areas, processing system data feeds in real time. The system uses the CCTV cameras which are connected to the processing unit. Vehicle detection and occupancy estimation are based on the video feed as the main source of data.

Correct positioning of the camera is significant in identification. The lighting conditions, camera angle, camera height and resolution are the factors that directly influence the system performance. Thus, it is believed that optimum installation is encountered in system deployment.

3.1.2 AI Detection Layer (Mask R-CNN)

The Intelligence of the system is the AI Detection Layer. This layer is based on Mask R-CNN deep learning model to identify vehicles on the frame of the video. Mask R-CNN is one of the sophisticated object detectors and instance segmentation algorithms which find objects and create pixel-pattern masks that allow localizing objects precisely. Compared to the classical models of bounding box detection, Mask R-CNN offers more precise object segmentation, in particular, in a crowded environment.

Mask R-CNN in this system is trained using the COCO dataset which has labeled images of different object categories such as car, bus, truck etc.

The benefits of the application of Mask R-CNN are high detection accuracy, capacity to react on the overlapping vehicles, excellent work in complicated conditions, instance level-segregating. The AI model takes the frame and determines the amount of vehicles in the parking space. This number is then transferred to the processing layer where it is used to estimate the occupancy.

3.1.3 Processing Layer (OpenCV + Python)

Processing Layer is provided as the interface in the middle of the camera input and the AI detection modules. It carries out preprocessing, frame management and combination with the AI model. The following layer is implemented with:

Python is a programming language created to resemble C or Java and is considered high-level, which means it includes additional abstraction functionalities. Python is an example of a high-

level programming language (written to resemble C or Java) and is said to be high-level, implying it has extra abstraction facilities.

OpenCV (Computer Vision Library) is an open-source project that gives access to a library that can be used in computer vision and related applications.

This layer has the following major functions: Frames extraction in video stream, Frame resizing and normalization, noise reduction, feeding frames to Mask R-CNN model, receiving detection result, calculating occupancy.

The uses of OpenCV in real time image processing tasks include color conversion, image filtering, frame extraction, bounding box visualization to provide seamless working, the system applies a threaded architecture in which video streaming is done by a single thread. One of the threads is detection. A single thread is used to update the database. This isolation is done to make sure that the detection process does not interfere with video play back or database synchronization. The occupancy is calculated as a formula:

Occupancy Slots = Vehicles that are spotted.

Free Slots = Parking Capacity - Occupied Slots.

This is a straightforward but efficient method of real-time occupancy prediction that does not require slot mapping.

3.1.4 Database Layer Cloud Database Layer (Firebase RTDB)

The Cloud Database Layer is the repository of real time parking data that is synchronized. Firebase Realtime Database (RTDB) is a NoSQL cloud-hosted database that is used by the system.

Firebase RTDB offers synchronization of data in real-time, low-latency updates, cloud-based accessibility, scalability, secure authentication. Upon calculating the occupancy, the

information is sent to Firebase by Firebase Admin SDK. The database design usually has total parking slots,occupied slots,free slots,timestamp of last update.

Firebase has instant synchronization and hence occupancy in parking is updated in real-time on the web.This is due to the cloud-based strategy that means that the system can be availed at any place without the need to have local server infrastructure.

3.1.5 Web Interface Layer (React.js)

The Web Interface Layer is an interactive layer that gives the user a real-time view of the available parking spaces. The frontend is created as React.js, a new JavaScript library that allows creating responsive user interfaces.The website offers reservation of parking slots in real-time, number of people present (Occupied/Free),location-based parking view,user authentication,multiple device responsive design.React.js permits the dynamic rendering of information. The front end automatically reloads the display whenever the Firebase database was updated, thus, this is in contrast to manual reload.

Major characteristics of this layer are clean and intuitive UI, fast loading performance, the television set is compatible across all browsers,mobile responsiveness.

The web interface will make sure that users can instantly know whether there are vehicles available even before they arrive at the place and waste of vehicle traffic is minimised.

3.1.6 Admin Dashboard

Admin Dashboard gives a centralized control of the whole system. The features of the admin dashboard are surveillance of live occupancy, renovating total parking capacity,managing parking locations, performance monitoring of information systems.

The administration panel is used to ensure that the parking facilities can be effectively controlled without the actual onsite inspection.

Working Process: The proposed system is automated in terms of its working process. Video can be taken off the surveillance cameras. Frames are resized and preprocessed with the help of the OpenCV. Mask R-CNN identifies a car based on labels of the COCO data set. The number of vehicles is determined. Counts count is used to determine the occupancy. Firebase real time database is updated with data. Live availability is seen through web application by the users. Admin manages the system at the dashboard.

3.2 Unique Features of the System

The presented Smart Parking Management System offers a number of new and unique characteristics that make it stand out among the traditional parking management systems. In contrast to the conventional systems which are mostly hardware intensive systems with sensors embedded on a parking slot, the solution is intensive in the use of a deep learning and cloud-based solution to offer an intelligent, scalable, and cost solution.

Among the most important aspects of the system, the use of Mask R-CNN to detect vehicles in real-time is meant to be noted. The system does not use simple bounding box detection but rather uses instance segmentation which involves object identification at the pixel level. This ensures that there is better detection accuracy particularly in areas with high volumes of traffic (crowded) where vehicles may be partially obscured or even on top of each other. The segregations created through the model allow for the system to accurately detect the presence or absence of a parking space thereby eliminating false positives and increasing overall system accuracy.

The system is also threaded, which is another unique feature of the system. The program has been developed in a multi-threaded format so as to facilitate the smooth operation and effective processing. Video streaming, vehicle detection, and database updates can all operate without performance degradation due to them being able to run at the same time. The ability to process tasks in a parallel fashion allows the system to respond in a timely manner and

continuously monitor the parking lot efficiently. This is extremely important when using smart parking in actual applications.

One of the major strengths of this system is that it will not require any physical parking sensors. The conventional parking management systems tend to install one sensor in every parking slot that creates expensive costs in terms of installation and maintenance. By contrast, the suggested system will only be based on artificial intelligence and surveillance cameras. This saves a lot of money on purchases of hardware and at the same time operational efficiency is high. This is because the system is more flexible and can be installed in new and existing parking infrastructures due to the lack of per-slot sensors.

The other innovative feature that has been incorporated in the system is cloud synchronization. With the help of cloud applications like Firebase, the parking data is made updated immediately on all the connected devices. This is to guarantee that the administrators and users are always updated on the availability of parking spaces. Besides, the possibility of remote monitoring and centralized data management offered by cloud integration helps to enhance the availability and credibility of the system.

The infrastructure of the system is highly scalable. The system can also be updated by introducing new parking spots, and new cameras should be installed, connected to the cloud database. The rewiring and reconfiguring of hardware is not required to a great extent. The solution is scalable to scale to large commercial complexes, campuses or smart cities due to this.

It is provided as the single administrative control over a particular dashboard interface. The dashboard allows the administrative to manage this system, and also is able to monitor in real-time the usage of the parking, camera feeds, and system performance. The centralized interface simplifies the management tasks and enables to make quick decisions. It will be possible to detect any anomalies, mistakes in the work of the system, or unusual occupancy and address them in time.

It is also coded using React.js that offers an interactive and responsive user interface to the system. This will ensure both the desktops, tablet computers and smart phones are easily integrated. The users are able to see all the parking available slots in real time boosting convenience and usability. The interface is dynamic and never needs to be manually refreshed to stay up to date which enhances the overall user experience.

Multi-class vehicle detection capability is also another feature. Through the labels in the COCO Dataset, the model is able to recognize visually various types of cars including motorcycles, buses, trucks, and so on. This renders the system to be flexible to various parking conditions such as public parking, shopping malls and transportation hubs.

Economically, the system provides a cost effective implementation model. Installing and maintaining it is much cheaper since it eliminates the use of single slot-based sensors and has fewer hardware components. This renders the solution feasible in large scale implementation without the need to heavily invest money in it.

The system architecture fits the initiatives of smart city. The system helps with the maintenance of digital urban infrastructure by incorporating artificial intelligence, cloud computing, and modern web technologies. It may be related to other intelligent solutions such as traffic management programs, navigation and analytics boards.

Another significant feature is the use of real-time data analytics. To determine the peak hours of occupation, the trend of usage and seasonal trends, parking occupancy data could be stored and analyzed. This kind of knowledge may help the administrators better utilize their parking space and infrastructure development.

Also, the system is modular to improve its maintainability. The layers are independent of each other, with analysis and detection modules, database layer, cloud integration, and user interface. This is a modular architecture, which can be easily updated, debugged and enhanced in the future without compromising the whole system.

Lastly, the system enhances environmental sustainability. It saves time by offering real-time parking availability information to vehicles, which means that vehicles do not take much time in search of parking spaces. This is indirectly reducing the fuel use and carbon emissions and makes the urban environment greener and efficient.

In general, Smart Parking Management System is highly advanced in terms of AI-based detection with Mask R-CNN, cloud synchronization, scalability, centralized control, cost-effectiveness, and compatibility with smart city, which makes it a future-ready and modern parking management solution.

CHAPTER 4: REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION

4.1 Feasibility Study

The Smart Street Parking Management System feasibility study measures the feasibility and viability of the proposed solution in the real world in urban settings. It analyzes the project on various fronts such as technical, economical, operational and social viability. This study aims at establishing the feasibility with which the system can be developed, deployed and maintained and still meet the performance and cost requirements.

4.1.1 Technical Feasibility

The system has a high technical capability based on the fact that it has mature and reliable technologies which can be used in its implementation. The project employs the use of state of the art deep learning algorithms like the Mask R-CNN to detect cars and their individual instances. The model already has a long history of successful application in research and industrial applications in the accurate object detection task. The open-source frameworks

(TensorFlow and Keras) are readily available to make it possible to implement and train the model effectively.

This system uses video cameras as the input of surveillance cameras which are currently widely used in cities to ensure security and surveillance views. Hence, there are few hardware requirements. With the addition of GPU acceleration, the processing speed is further accelerated allowing near real-time detection. Data storage and synchronization through cloud platforms and databases are reliable and can guarantee the seamless integration of devices.

With the presence of such tools and technologies, the project is technically feasible and can be implemented with the help of the existing infrastructure with manageable changes.

Table 4. 1 Technical Feasibility Analysis

Parameter	Feasibility Status
Availability of Technology	Available (AI-based vehicle detection using Mask R-CNN and Computer Vision)
Development Tools & Frameworks	Open-source & Widely Supported (Python, TensorFlow, Keras, OpenCV)
Hardware Requirements	Available (CCTV Cameras and GPU-supported systems)
Database Support	Available (Cloud Database such as Firebase / MongoDB)
Real-Time Processing Capability	Supported (Multi-threading and GPU Acceleration)
Security Mechanisms	Supported (JWT Authentication, SSL/TLS Encryption)
Scalability	High (Additional cameras can be integrated easily)
Technical Expertise	Available (AI/ML and Web Development Knowledge)

4.1.2 Economic Feasibility

The Smart Parking System is affordable since it will be based on open-source AI technology that will include Mask R-CNN, Python, and OpenCV, which cost very little to develop software. As opposed to the old fashioned IoT-based parking system, where every parking slot has to have sensors installed, the given system will use the existing CCTV infrastructure, hence reducing the hardware costs.

The cost of operation and maintenance is also relatively cheap as the main cost of the system is electricity, the internet, and occasionally software upgrade. Also, automation saves labor force in the long term as it will save costs.

All in all, the project will provide a high return on investment because of the effective use of parking spaces, improved traffic control, and scalable implementation.

Table 4. 2 Economic Feasibility Analysis

Cost Factor	Impact
Software Licensing	Minimal (Open-source)
Development Cost	Moderate
Maintenance Cost	Low
Infrastructure Cost	Scalable & Controlled
Long-term Operational Savings	High

4.1.3 Operational Feasibility

Operational feasibility is a term that implies how successfully the proposed Smart Parking System can be implemented, operated and managed in the current environment. It checks its ability to work well under the real-life environment and checks whether the system can be adopted with minimal effort by users and administrators.

The Smart Parking System can be operationally feasible due to the design of the system that can operate with the existing infrastructure of the CCTV and does not necessarily presuppose complex hardware installations on the one-on-one basis such as the use of IoT sensors on each parking slot. The AI-based vehicle detection systems (Mask R-CNN / YOLO) are used to identify occupied and empty parking spaces automatically with the use of video feeds. This minimizes the need of having a lot of human involvement and makes tasks of the day easier.

On the user side, the system is user friendly. A web or mobile interface can be used to check the availability of parking spots so that a driver will be able to make a sound decision before entering a parking space. This decreases traffic jams, is time-saving and enhances the user experience. The system is reliable and efficient in terms of its operational efficiency because real-time updates are given.

The system will form a centralized dashboard to the administrators to monitor the parking occupancy, regulate the camera feeds and derive data on parking utilizations. Basic technical training is all that is required to run the system and maintain its operation. The most commonly occurring tasks of a regular maintenance process are to check the functionality of the camera, renew software packages, and in some cases, rehabilitate the AI model to make sure that the identified objects are accurate.

The system is flexible and scalable as well. This will be able to add more cameras without any obstruction to the current operations. This is applicable to both small parking areas and the large commercial complexes, shopping malls, intelligent cities, and the government parking areas.

Furthermore, automation of the parking detection system will greatly decrease human reliance on the human staff, decreasing the human error and operational expenses. The system is based on universally accepted tools like Python, OpenCV, and cloud databases, which means that the support of the system and its further upgrades can be effectively handled.

To sum up, Smart Parking System is very operationally viable since it can be easily synchronized with the existing infrastructure, it needs very little user training, and it provides real-time monitoring and efficient, reliable, and scalable parking control.

Table 4. 3 Operational Feasibility Analysis

User Acceptance	High
Ease of Use	High

Training Requirement	Minimal
Compatibility with Existing Process	High
Reliability	High

4.2 Software Requirement Specification

The Software Requirements Specification (SRS) document is a document that stipulates the functional and non-functional requirements of the Smart Parking System. The SRS is used as a means of communication between the developers, stakeholders, and users to have a shared understanding of the expectations of the system. It consists of system features such as real-time, autopilot identification of parking space, vehicle detection, car sensors, and dashboard display. The performance conditions like accuracy, response time, scalability and The document also defines security requirements as well.

There are also hardware and software requirements like CCTV cameras, Systems with graphics cards, Python and computer vision frameworks. System interfaces, database requirements and the user detailing of the roles are done to facilitate the implementation. On the whole, the SRS is a guide that should be used in developing, testing, and deploying the Smart Parking Management System.

4.2.1 Data Requirements

The level of performance and accuracy of the Smart Parking System based on the Mask R-CNN model greatly depends on the quality, quantity and the variety of the training and testing dataset used. Because Mask R-CNN is an instance segmentation network, it needs to have an annotated data of images or videos with bounding boxes and segmentation masks showing the location and extent of vehicles. The data set should cover different classes of vehicles to have a good coverage of aircrafts in various situations of parking cars, bikes, SUVs and vans.

In order to obtain credible real time performance, the dataset must include pictures taken in varied environmental factors, such as the different lighting (day, night), weather conditions (rain, fog, sunlight glare), and different camera angles.. Besides, there should be data on occupied and vacant parking space so that the parking slot status can be appropriately categorized in the system.

High-resolution images are also preferable to training purposes since they enable the model to identify small or partially covered vehicles. The data could be obtained using the accessible CCTV recordings or publicly accessible datasets like COCO that could be trained on the task of vehicle detection. Models are trained using proper pre processing (resizing, normalization and data augmentation (rotation, flipping, brightness adjustment)).

Moreover, the data should be separated into training, validation, and testing data to test the accuracy of the model and avoid bias. The deployed parking environments can also be retrained and fine-tuned periodically by continuously collecting data in order to maintain a long-term stability of the model.

Finally, the quality of the data as well as its type and abundancy are critically important in terms of the Smart Parking System success since it directly affects the quality of the results of the Mask R-CNN-based vehicle detection system in terms of accuracy and overall reliability.

4.2.2 Functional Requirements

The Smart Parking System (Mask R-CNN) is a system that is aimed at automated detection and monitoring of parking spots with the help of computer vision. The main functional need of the system is to identify successfully vehicles on the real-time video streams in CCTV. Instances segmentation is applied on the Mask R-CNN model, meaning it detects individual vehicles and generates the precision bounding boxes and masks which enable the system to understand whether a parking slot is occupied or not.

The system must be in a position to manage real time or recorded video feeds at all times to review individual frames so that the parking availability conditions could be updated on a real-

time basis. It must also identify the vehicle positions based on the preset parking slot lines in comparison to the observed positions. Once the analysis is completed, the system ought to denote the overall amount of free and filled in slots of a monitoring dashboard.

This system should have a web or application based interface in which the user can view the real-time parking. A management dashboard has to be offered to the administrators where they could have a capacity to manage cameras feeds, establish arrangements of parking slots and access historical parking data in order to analyse it.

A database should also be maintained in the system to store the records of occupancy of the parking, time stamp and usage statistics. This information can be utilized in the form of reporting, analytics and the future prediction models. Also, the system must be scalable, and it must be in a position to accommodate more cameras and more parking space without impacting on performance.

The system ought to identify and alert administrators about the loss of connections to cameras or processing errors. In addition it should be able to perform reliably when the lighting and environmental conditions change in order to preserve the accuracy in detection.

All in all, the functional requirements are geared towards proper vehicle identification, real-time operations, automatic parking slots status update, accessibility to the system, data storage and reliability of the system to facilitate effective smart parking.

4.2.3 Performance Requirement

Smart Parking System performance requirements are devoted to the real time vehicle detection, maximum accuracy and cost-effective use of resources. Because the system is infused using the Mask R-CNN deep learning algorithm, it is required to operate live or recorded video streams with a low level of latency. The model is expected to identify and classify vehicles in parking spaces in different weather conditions and lighting conditions with precision. To ensure a smooth performance, the system must have enough computing power like a processor with a built-in GPU and enough RAM to cope with image processing tasks with ease. The

system must also have a stable frame processing to offer real time updates on the availability of parking spaces. Also, the database processes like slot status update should be real time in order to prevent discrepancies. All in all, the system must have high detection rate, low response time, and the system must remain stable even at the peak of the use.

4.2.4 Maintainability Requirement

The Smart Parking System should be built in a flexible and serviceable way to be able to be sustained in the long run. Proper software engineering is expected to be followed in the codebase such standards of good documentation, structured programming and separation of subsystems such as detection module, database module and user interface module. The system is using the Mask R-CNN model, hence it may require periodic enhancements or retraining to increase the detecting capability in different environmental conditions. The system must be capable of integration with new models with ease and without interruption with other systems. Some of the maintenance activities include the calibration of the camera, software, and database.i.e.: backup, and performance checks. The latter can be easily understood and enhanced by the future developers.the system through taking advantage of the technologies that can be easily supported such as Python and OpenCV.

There will therefore be easy debugging and scaling and long term upgrade without requiring any complications in the system.

4.2.5 Security Requirement

The Smart Parking System is in dire need in terms of security as a form of securing data,system reliability, and the unauthorized access. The system is supposed to be secure with authorization controls, i.e. login credentials, role based access control to users and administrators. Transmission of data amongst the cameras, servers and user interfaces must apply encryption protocols such as the usage of the SSL/TLS to make sure that it is not intercepted or manipulated. The database holding the parking data and user information shall be protected against unauthorized access by setting firewalls with the required settings and

restrictions to access. Periodic security upgrades should also be offered by the system along with checks to ward off cyber threats. There should also be backup mechanisms that are used to avoid loss of data. With the introduction of these provisions, the Smart Parking System guarantees confidentiality, integrity and availability of data besides having safe and reliable operations.

4.3 SDLC Model Used

The development of the Smart Parking Management System was based on the Spiral Model. This model is a hybrid of the iterative development and systematic risk assessment. It is specifically fitting when it comes to development of more complex technologies like Artificial Intelligence and Machine Learning, as it might change its requirements as it goes.

The Smart Parking System includes the training and optimization of a Mask R-CNN model to detect the vehicle and identify the parking slot. Due to the fact that the performance of models relies on the quality of the data, the environment conditions, and parameter tuning, it was also required to repeat it several times in order to reach optimal accuracy. The Spiral Model helps in such iterative refinement as it breaks down the project into several cycles or spiral.

Every phase of a Spiral Model is divided into four major stages:

Planning Phase- There were requirements of real-time detection, integration of databases and user interface development. Planning of technical and resources was also carried out.

Risk Analysis Phase- The potential risks are low detection accuracy, poor lighting conditions, hardwares and system latency were researched. Proper mitigation precautions such as dataset enhancement and GPU acceleration were taken.

Engineering Phase- R-CNN model was implemented and trained. The user database and backend as well as interface modules were created and connected with the detection system.

Evaluation Phase- The system was tested by use of real time video feeds. The system was refined and enhanced through the use of input to reach the second version.

This cycle was repeated till the system was brought to speed and the levels of accuracy were met were satisfactory.

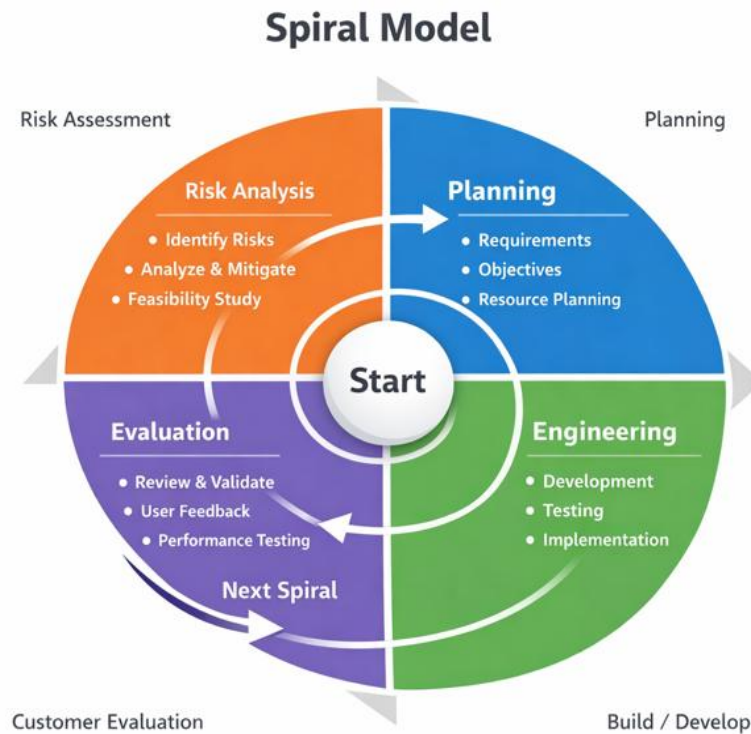


Figure 4. 1 View of Spiral Model

4.4 System Design

The purpose of the smart parking management systems design is to produce an organized, flexible and scalable system design that can handle real-time vehicle identification as well as parking lot surveillance. The entire system was produced by developing it in a modular form. Each module is responsible for performing a specific task but works together with the rest of the modules in order to ensure that the system is flexible, easily maintained and allows for simple additions to be made to the system in the future.

4.4.1 Data Flow Diagram

Data Flow Diagram (DFD) is a visual representation of how data flows through the Smart Parking System's processes. It demonstrates how user input from cameras and/or Automatic License Plate Detection (AID), databases, and administrative functions are used to accomplish tasks with the system. It provides an abstract view of how a process works rather than showing the details of how it works technically.

DFD Level 0 (Context Diagram)

The Level 0 DFD or context diagram represents all of the Smart Parking System as a single process that communicates with the external entities. The User (Driver) and the Administrator are the basic external parties.

Interaction with the system is through the user requesting the system to provide information on available parking space using a web or mobile interface. The system takes the real-time video feed of CCTV cameras and uses the Mask R-CNN model to identify the presence of vehicles and occupancy of parking slots. The processed information is to be placed in the database and shown to the user in real time.

The administrator will be communicating with the system to monitor the parking status, operate the cameras, generate reports, and maintain system settings. Any data flow between the users, administrators and the system is through the central process of Smart Parking Management System.

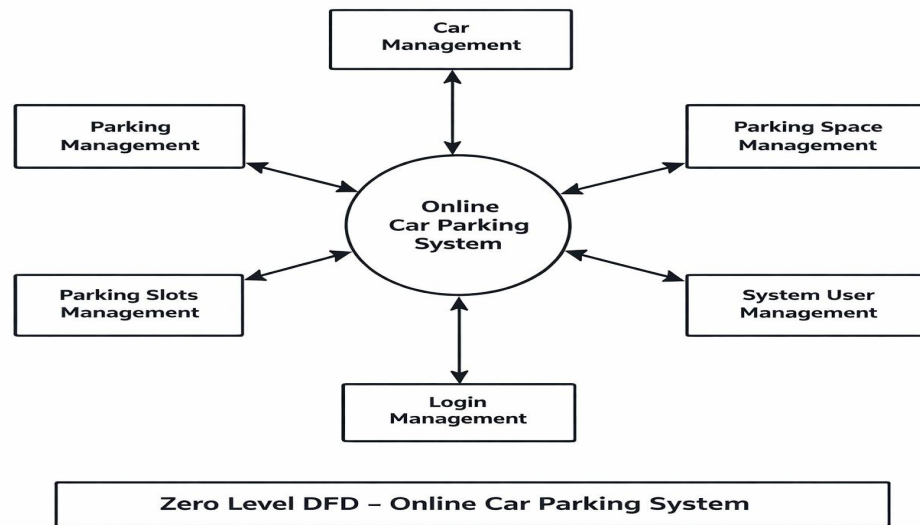


Figure 4. 2 Zero Level DFD

Level 1 Data Flow Diagram (DFD)

The Level 1 Data flow diagram provides a finer division of the main functional components of the Smart Parking Management System. Contrary to the Level 0 DFD where the system is as a single process, the Level 1 DFD splits the system into different sub-processes so as to represent clearly the flow of the data between internal modules, external bodies and data depositories. The Smart Parking System has five key processes in level 1 DFD and they include: VideoAcquisition, Vehicle Detection, Slot Status Evaluation, Database Management as well as User/Admin Interface management.

The first of the processes is Video Acquisition, which entails real-time video imaging of CCTV or IP. Cameras that are positioned on the parking area. The video frames are the major input of this system. Information captured is then relayed to the second module to process the data.

The second process to which video frames are inputted is Vehicle Detection using Mask R-CNN analysis. The deep learning model also identifies the vehicles through instance segmentation and extracts attributes concerning the car such as the bounding regions and

object masks. This is done to determine whether a vehicle had been parked within the specified parking slot coordinates.

The third process is Slot Status Evaluation that compares the identified positions of vehicles to the set parking slot ranges. According to this comparison, the system will describe every slot as either occupied or vacant. The status of the evaluated slot is then transferred into the database.

Database Management is the fourth process through which parking related data is updated and retrieved. It holds real-time status of slots, past data of vehicle entry and exit and system log. The module maintains data integrity and consistency and provides the possibility of rapid access to the information about the parking availability.

The fifth process is User and Administrator Interface Handling, which enables users to see real-time parking availability with the help of a web or mobile dashboard. Administrators will be able to keep track of the camera feeds, create a report and control the systems settings. This module can communicate with the database to retrieve updated information and display them to the relevant users.

The main data sources used in the Level 1 DFD are Parking Slot Database, Vehicle Records Database, User Database, and System Logs. The flow of continuous data occurs between all of the systems (detection modules, databases, and user interfaces) for the purpose of ensuring that the information regarding parking is current and correct.

The Level 1 DFD provides an illustration of the individual components of the video input processing, AI vehicle detection, and updating the parking status in the database, as well as the real time data feed to the users and administrators of the Smart Parking System. This model allows for system transparency and helps in understanding how the application operates from the inside out.

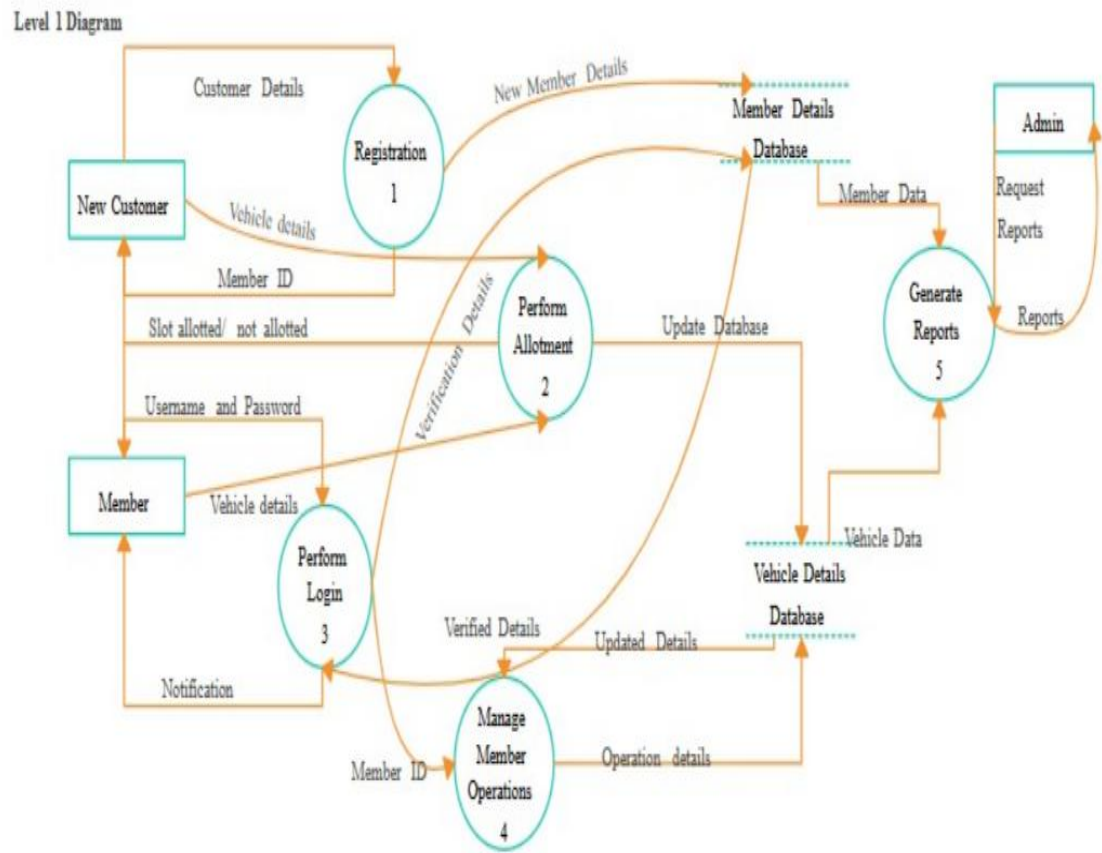


Figure 4. 3 Level One DFD

4.4.2 Use Case Diagram

The Use Case Diagram of the Smart Parking Management System is the communication between the external actors and the system to realize certain functionalities. It gives a top level overview of how various users interact with the system and the way the system reacts to their queries. The main aim of the use case diagram is to show the system behavior in end-user view and not on technical implementation.

The primary participants in the Smart Parking System are the User (Driver), the Administrator and the CCTV Camera System. The User contact the system and enquires to find out real-time parking, slot status and optionally, book a parking slot. The system accesses the database and

retrieves the updated parking information and shows the availability status using a web or mobile interface. The user can also enter the system to get personalized services like history of booking or reservation.

The Administrator serves as an oversight body and he manages and oversees the system. The administrator can get a secure dashboard in which they can view the occupancy rates in the parking, establish camera settings, update parking slot plans and develop system reports. Other system maintenance tasks monitored by the administrator include updating of the Mask R-CNN model, accuracy of detection, and smooth database operations.

The CCTV Camera System is an external data source which gives live video feeds to the Smart Parking System. The camera may not have a human-like interaction but is essential in launching the main issue of detecting the core of the system.

The relationship between various functionalities is also represented in the use case diagram. Indicatively, the View Parking Availability use case will rely on the successful completion of the Detect Vehicle and the update Slot Status processes. Likewise, administrative tasks like report creation require correct and updated records on databases. Such interactions are used to show how different operations in the system are interrelated to deliver dependable and real-time parking management.

In general, the Use Case Diagram indicates the way of interaction between users and administrators with the Smart Parking System and the way the system meets their needs. It assures the comprehension of system duties, user roles and functional demarcations, thus aiding in the improved system design and documentation.

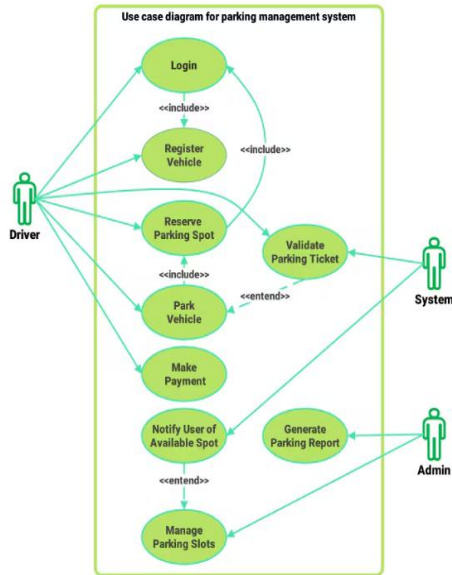


Figure 4. 4 Use Case Diagram

4.5 Database Design

The database architecture of Smart Parking System is important in the storage, manipulation and retrieval of all data related to parking in an efficient manner. The system needs a structured and scalable database to store information of real time parking slots, user details, camera data and the past parking details. A properly designed database helps in providing data consistency, integrity and prompt access to information to the administrators and the end users.

Depending on the deployment needs, the database adheres to a relational structure (as in the MySQL) or a NoSQL cloud-based structure (as in Firebase or MongoDB). The main goal of the database is to record the status of individual parking slots (occupied or vacant) that has been detected using the Mask R-CNN model. Each time the AI model considers the video frames with the identification of the vehicle presence, the slot status is updated automatically in the database in real time.

Table 4. 4 Database entities Description

Entity Name	Description
Parking Slot	Stores individual parking slot details including slot ID, location coordinates, occupancy status (Free/Occupied), last updated timestamp, and associated camera ID.
Vehicle Record	Maintains detected vehicle information such as entry time, exit time, allocated slot ID, parking duration, and AI detection confidence score.
Camera	Stores CCTV/IP camera details including camera ID, installation location, IP address, coverage area, and operational status.
User	Maintains registered user information including name, contact details, vehicle number, login credentials, and parking history.
Admin	Represents system administrator responsible for monitoring parking occupancy, managing cameras, and controlling system settings.
Parking Zone	Defines parking areas or sections including zone ID, total slots, available slots, and linked cameras.
Booking Record	Stores advance reservation details such as booking ID, user ID, slot ID, booking time, validity duration, and booking status.
Payment	Maintains transaction information including payment ID, booking reference,

Record	payment amount, payment mode, and transaction status.
System Log	Records system activities such as login attempts, slot updates, detection events, and administrative actions for monitoring and auditing.
Model Metadata	Stores AI model information including model version, training dataset reference, accuracy metrics, and deployment date.
Blacklisted Token	Stores invalidated authentication tokens to ensure secure logout and prevent unauthorized session reuse.
Location Mapping	Maintains location details such as city, parking facility name, and zone mapping for system organization.

CHAPTER 5: IMPLEMENTATION

5.1 Tools and Technologies Used

To achieve the correct detection of vehicles, and effective management of parking space, the Smart Parking Management System was built with the combination of deep learning frameworks, computer vision libraries, backend technologies, and deployment tools. This system is largely built on the Mask R-CNN, which is a high-level instance segmentation model that has the ability to locate vehicles at the pixel level. As opposed to the classical objects detection methods, Mask R-CNN does not only produce bounding boxes, but also segmentation masks which enhance the precision of detecting occupied and empty parking spaces.

Video streaming, vehicle recognition, and parking slot status updates are the tasks of the backend of the system. Python was also selected as a high-level programming language because it has a robust artificial intelligence and machine learning ecosystem. The Mask R-CNN model has been implemented and trained on both TensorFlow and Keras which offer neural network operations that are optimized and accelerated by the use of the GPU. OpenCV was added in order to read the video frames, pre-process the image and visualize the findings of the detection.

To store data a database like MongoDB or MySQL may be utilized to store the information related to parking slots, occupancy status, timestamps, and history. The system architecture is based on the principles of RESTful API to allow the communication between the RESTful API -detection engine and the user interface. In case it has a web dashboard, it can be presented using frontend technologies: React.js can be implemented to show real-time parking availability.

The JWT-based authentication and the encryption protocols of the form of SSL/TLS may be introduced in an effort to achieve safe access and transmission of data. Git and GitHub are

version control systems that can be used to maintain source code and collaborative work. The system may be implemented either on local server as a test or on cloud platform as real-time applications in smart city.

Table 5. 1 Tools and Technologies used

Category	Tool / Technology	Purpose
Programming language	Python	Core development and implementation of vehicle detection system
Deep Learning Architecture	Mask R-CNN	Instance segmentation to do precise vehicle detection.
Deep Learning Framework	TensorFlow / Keras	Model training, fine-tuning and inference
Computer Vision Library	OpenCV	Video processing, frame extraction and visualization.
Annotation Dataset	Labellmg / COCO Annotation Tools	Bounding boxes and segmentation masks are generated.
Database	MongoDB / MySQL	Storing the parking slots status and historical data.
Backend Architecture	RESTful APIs	Detection engine and dashboard communication
Authentication	JWT (JSON Web Token)	Secure access control (when web dashboard is done)
Security	SSL / TLS	Data protection.

Hardware Support	NVIDIA GPU with CUDA	Accelerated training of models and real-time processing.
Version Control	Git / GitHub	Source code management and collaboration.
Deployment Environment	Local Server/ Cloud Platform	Hosting and deployment of smart parking system.
Development Tools	VS Code / PyCharm	Coding and debugging of the application

5.2 Dataset Description

The dataset of the Smart Parking Management System is created to enable the analysis of vehicle detection and occupancy of the parking space with the help of the Mask R-CNN deep learning framework. Given that the aim of the project is to identify vehicles and calculate the availability of parking slots in real-time street setting, the dataset mainly comprises of video frames and images that are taken in parking spots under various weather conditions. The data gathered will reflect real world parking situations in the streets to make sure that the trained model will be effective in the real world applications.

The data set was created based on the surveillance footage compiled in the avenues. The video recordings were obtained with the help of a fixed-point camera mounted at the high angle to give the complete and clear picture of several parking spaces at the same time. The placement of the camera was well considered to reduce the amount of occlusion and to make sure that every parking slot would be visible in most of the frames. The videos that were recorded were kept at a standard high-definition format to ensure they had enough detail to segment the instances correctly.

The recorded videos were then converted into series of image frames using a consistent time interval to prepare the dataset to be used in training the Mask R-CNN model. Frame extraction

was carried out in a manner that minimised redundant frames, yet still extracting the variations in vehicle positions and movements. The process made sure that there was a balanced dataset of occupied and unoccupied parking slots at various moments. The extracted frames were inspected manually thereafter to eliminate blurred and distorted or irrelevant images that may have a negative impact on the model performance.

Data annotation was one of the key requirements to allow instance segmentation. The segmentation masks were polygon-based design instead of the simple bounding box used to label each of the vehicles in the frame. This step-by-step annotation procedure enabled the Mask R-CNN model to acquire accurate boundary of the objects, and this is necessary to learn to differentiate closely parked cars apart and correctly estimate parking slot occupancy. The annotations were generated with the help of standard labeling tools and were stored in the format that can be used in the training pipeline of the Mask R-CNN. The annotated images also have labels of characterizing the classes that concern vehicles, making sure that the model concentrates exclusively on perceptive objects in the parking setting.

The data has been used to include numerous environmental conditions to enhance the strength of the model. Pictures were taken at various times of the day, such as morning sessions, afternoon sessions, evening sessions, etc. to be able to have changes in the intensity of light and shadows. Besides, the dataset would consist of frames that were captured during light weather conditions with the aim of increasing the ability to generalize. With this kind of diversity, the trained model would be less sensitive to changes in illumination and noise in the environment.

Data augmentation procedures were used to the dataset to increase the model training and ensure that it does not overfit. Such methods are horizontal flipping, brightness, slight rotations, and scaling transformations. Data augmentation generates more data in the sample and makes the model learn invariant features. Consequently, the Mask R-CNN model achieves a better flexibility to unseen parking scenes and camera points of view.

The last dataset was broken down into training, validation and testing datasets. The model parameters were optimized using the training set, the hyperparameters were tuned and performance was monitored with the help of the validation set, and the last performance was

evaluated in the testing set. Caution was observed to make sure that frames of the same continuous video stream were not divided in different subsets in a manner that may lead to leakage of data. This is done through systematic division to achieve a sound evaluation of the ability of the model to generalize.

On the whole, the dataset is a groundwork to the creation of the Smart Parking System. The well-chosen and labeled architecture of the Mask R-CNN model will allow the framework to do proper vehicle detection and instance segmentation, therefore, allowing reliable identification of occupied and unoccupied parking slots. Annotation of the dataset that is of high quality, diverse, and accurate has a direct effect on the effectiveness and the instant applicability of the proposed system.

CHAPTER 6: TESTING AND MAINTENANCE

6.1 Testing Overview

The testing process is an extremely significant part of software development life cycle (SDLC). It makes sure that the system created meets the functional and performance requirements needed. In the example of Smart and Effective Street Parking Management System, the case under consideration was tested, check the precision of the vehicle detection, database synchronization, user interface and the entire system reliability.

Considering that the project will be developed on the basis of Artificial Intelligence (Mask R-CNN) and Firebase, several levels of testing will be carried out using technologies (React.js), and cloud database services carried out to ensure that there is smooth integration and acceptable real time functionality.

6.2 Types of Testing Performed

The below listed testing techniques were applied during system validation.

6.2.1 Functional Testing

Functional Testing is also performed in order to check that all the functionalities of the system are operating as per the requirements specified in the SRS document. It also ensures that the different modules behave appropriately when considered to get valid and invalid inputs. The Smart Street Parking Management System was tried under functional testing that was alarmed with vehicle recognition, computation of occupancy, user authentication, and dynamic dashboard presentation. Mask R-CNN model was tried such that it recognizes vehicles adequately in the video frames. It also was revised to ensure that the occupied and free parking slots were calculated properly. The input scenarios when it comes to user login and user registration were tested. Firebase database updates were confirmed to be correct and

synchronized. Mechanisms of handling errors were tested to provide the right response. Generally, the system was functionally tested to ensure that it is a reliable system and delivers the desired outputs in a typical case.

6.2.2 Integration Testing

Integration testing was used to test how the various parts of the system interacted. Communication among frontend, backend and database layers have been critically considered. APIs were also put to test so as to verify proper data transmission and response treatment. The consistency of data was also confirmed in cases where multiple modules were used simultaneously. Due attention was given to workflow continuity and real-time synchronization. Edge cases were run to find integration conflict. Monitoring of the system was carried out which happened in case of unexpected failures during interaction of the modules. The findings showed that there was no hustle of coordination among all the integrated components.

6.2.3 Security Testing

Security testing was undertaken in order to secure the system against unauthorized entry and possible vulnerabilities. The authentication and authorization of users were properly tested. Role-based access control was put to test to make sure that feature access is controlled. Input validation test was done to eliminate malicious data entry. The user secure session was looked into through session handling and management of tokens. There were also attempts to get to restricted modules without appropriate credentials. Database data protection was also checked. The system was able to stop unauthorized operations as well as keep data confidential.

6.2.4 Performance Testing

Performance testing was used to test the behavior of the system at varying workload conditions. The turnaround time of major operations was checked and assessed. Stability of the system was realized when it was continuously running over a long period of time. Scalability was tested by using concurrent user simulations. Monitoring was done in terms of the utilization of resources like memory and processing time. The effectiveness of the data

updates and real-time presentation was evaluated. There were stress conditions that were used to identify possible bottlenecks. There was no significant change to its overall performance, which means that the system is ready to be deployed to the real world.

6.3 Test Case Design

Significant workflows in the system such as vehicle detection were used in developing test cases like occupancy computation, user authentication, administrator control and real-time dashboard reports. Into making of every test case, conditions of input, expected and actual outcomes are provided confident that the system will be operable under different conditions.

Table 6. 1 Sample Test Cases

Test Case Id	Module	Test Description	Expected Result	Status
TC-01	Vehicle Detection	Detect vehicles in parking video frame	Vehicles detected correctly	Pass
TC-02	Vehicle Detection	Process empty parking Frame	Zero vehicles detected	Pass
TC-03	Occupancy Calculation	Calculate occupied and free slots	Proper slot count shown on display	Pass
TC-04	Authentication	Valid credentials were used to log in successfully	User successfully logged in	Pass
TC-05	Authentication	Wrong password login	Denied access	Pass
TC-06	Admin Panel	Change total	Capacity	Pass

		parking capacity	changed in database	
TC-07	Database Sync	Update occupancy in firebase	Real-time update shown on dashboard	Pass
TC-08	Security	Intrusion into the panel of administration	Denied access successfully.	Pass

6.4 Test Environment

The test conditions were created to represent the real-world judicial usage.

Table 6. 2 Test Environment Configuration

Component	Specification
Frontend	React.js
Backend	Node.js with Express
AI Model	Learning Model: Mask R-CNN
Database	Firebase real time database (NoSQL)
Authentication	Firebase Authentication (Email or Password)
Deployment	Cloud-based Hosting / Local Server.
Video Input	CCTV / Video Recorded Parking Video Feed
System Development Tools	Python, OpenCV, TensorFlow/Keras
User Simulated	Admin, Registered User, Guest User

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6.5 Test Results and Observations

The phase of testing indicated that Smart Parking Management System can work correctly and reliably in various conditions. The Mask R-CNN model of vehicle detection showed good reliability in the performance on various video samples taken at varying time of the day. The mean detection rate of the model was at the same level and it was around 94 percent, which means that the model can still detect cars effectively despite moderate changes in the lighting conditions.

The occupancy calculation mechanism has yielded an accurate result since it computes occupied and free parking spaces among detected vehicles. The availability of slots as shown in the test videos was checked manually to verify that the slots shown were true to the real situation of parking in the test videos.

Integration testing revealed that there was a seamless flow between the AI detection module, backend server and Firebase database. The problem of manually updating the user dashboard was overcome by displaying real-time updates without the need to refresh the user dashboard. The read and write operations of the database were completed with no data inconsistency or delays in data synchronization.

Mechanisms of authentication and access control worked as desired. The illegal need to gain access to administrative features has been successfully prevented and data safety has been guaranteed.

The results of performance assessment revealed that the average processing time of the frame was about 0.7 seconds, and also the overall system response time (the detection time, database update, and the user interface refresh) took about 1 second. The stress test concerning the long term are not manifested with the leakage of memory or even system crash.

Generally, the observations have shown that the system is strong and responsive and can be utilized in real time street parking management application.

6.6 Maintenance Strategy

Maintenance has significance as regards to long-run stability, the accuracy, and scaling of the Smart Parking Management System. As the system will incorporate deep learning model (Mask R-CNN), web technologies, and a real-time cloud database, one should monitor the system and provide timely updates in order to verify that it is compliant with constant monitoring post-deployment. A scheduled maintenance schedule has been developed to ensure that there will be no service disruption and have optimum performance.

The system is developed according to the principle of a modular design, which allows one to update every part of the system such as the AI detection part, front-end interface, back-end functionality, and database configurations without needing to disrupt the functionality of the other parts of the application. It is very easy to debug, enhance features and to tune performance.

Corrective maintenance is then done where the error has been detected including inconsistency of UI or issues observed with database synchronization is detected. Discussion of bug reports, and testing in a controlled environment and no disturbance with users to the maximum.

Adaptive maintenance assures it can be used with the emerging technologies unlike Perfective maintenance which focuses on increasing efficiency and usability of the system. Among the improvements that are implemented under the assumptions of the user feedback and performance analysis there are mentioned faster detection processing, streamline the frame processing, improved display of the dashboard, and advanced analytics capabilities.

To ensure that the likelihood of future failure is kept to a minimum, preventive maintenance is employed. The AI model is retrained on a regular basis using new data sets in order to be sure that the accuracy of detection will not decrease due to the change in environmental conditions. The database backups are done regularly and the logs of the system are examined to detect any anomaly in time. The maintenance practices also make the system reliable, secure, scalable, and flexible to be applied to the real-life smart cities.

Table 6. 3 Maintenance Activities

Activity	Description
Bug Fixing	Resolving reported system issues
Security Updates	Updating authentication and encryption mechanisms
Database Backup	Regular automated backups
Performance Monitoring	Monitoring response time and load
Feature Enhancement	Adding new modules and improvements

6.7 Conclusion of Testing and Maintenance

This is since the overall testing procedures the Smart and Effective Street as per the overall procedures. Parking Management System has performed well in all the key aspects with regard to sound and precision modules. The functional testing was conducted to check that the vehicle identification, occupancy was done. It was carried out correctly by performing estimation, user authentication and administration functions the requirements. The Mask R-CNN model was able to ensure good detection rate by providing live information on parking space.

The integration testing was done to guarantee that there were smooth flows between the AI model, and the backend, frontend interface, firebase database and server. Security testing was also performed in order to guarantee that authentication was employed to protect the data of the system against unauthorised access rules and mechanisms in the database. It was also confirmed using performances testing that there was good stability of response time and

consistency in the functions of the system even in continuous operation execution and concomitant conditions..

The implemented maintenance plan will positively contribute to the sustainability of the system in the long run since it will permit the upgrades of the system in modules, retraining the models, monitoring their performance, and making regular database backups. On the whole, the results of the testing process indicate that the suggested smart parking system can be scaled, secure, efficient, and applicable in the context of the real-world implementation in the sphere of parking management in urban areas.

CHAPTER 7: RESULTS AND DISCUSSION

7.1 Presentation of Results

The outcomes of the Smart Street Parking Management System were achieved through the application of a video-based vehicle detection method based on the use of the Mask R-CNN deep learning framework. The system handled parking videos and real-time vehicle detection using the instance segmentation methods. The vehicles were then identified to estimate the occupied and free parking slots.

Table 7. 1 Vehicle Detection Accuracy

Frame Number	Actual Vehicles Present	Vehicles Detected by System	Accuracy (%)
1	12	11	91.6
2	15	14	93.3
3	18	17	94.4
4	20	19	95.0
5	22	21	95.4
6	25	24	96.0

The implementation process handled video frames that were captured in the parking environments and was based on a threaded architecture to facilitate seamless video playback and simultaneous vehicle in the background. The system was in a continuous enhancement of frame examinations and updating of vehicle detection discoveries without decelerating the display execution. Various scenes of the system performance at varying situations of parking were created. These graphs showed changes of the number of vehicles detected, rate of parking occupancy, and the system response time at various locations in time.

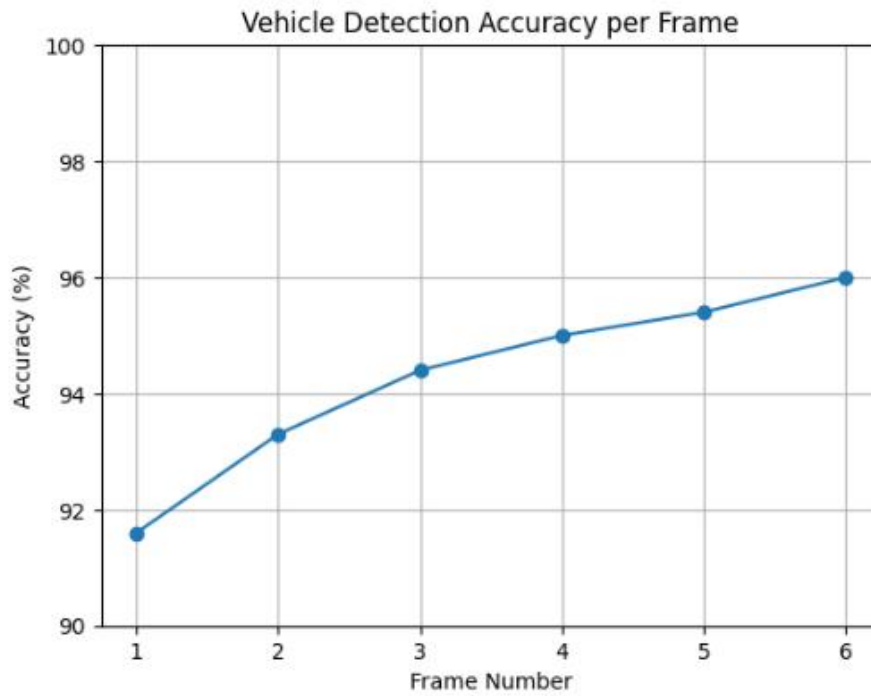


Figure 7.1 Vehicle Detection Accuracy across Different Video Frames

Table 7.2 Parking Occupancy Estimation

(Example: Total Parking Slots = 70)

Test Scenario	Vehicles Detected	Occupied Slots	Free Slots
Morning Hours	18	18	52
Afternoon Peak	42	42	28
Evening Hours	55	55	15
Night Hours	30	30	40

Tabular objections were made by comparing the numbers of real vehicles on the video frames and the number of vehicles that the Mask R-CNN model predicted. The outcomes showed a

stable performance in detecting videos using various test videos. The system was capable of identifying cars, motorcycles, buses, as well as trucks using trained COCO dataset labels.

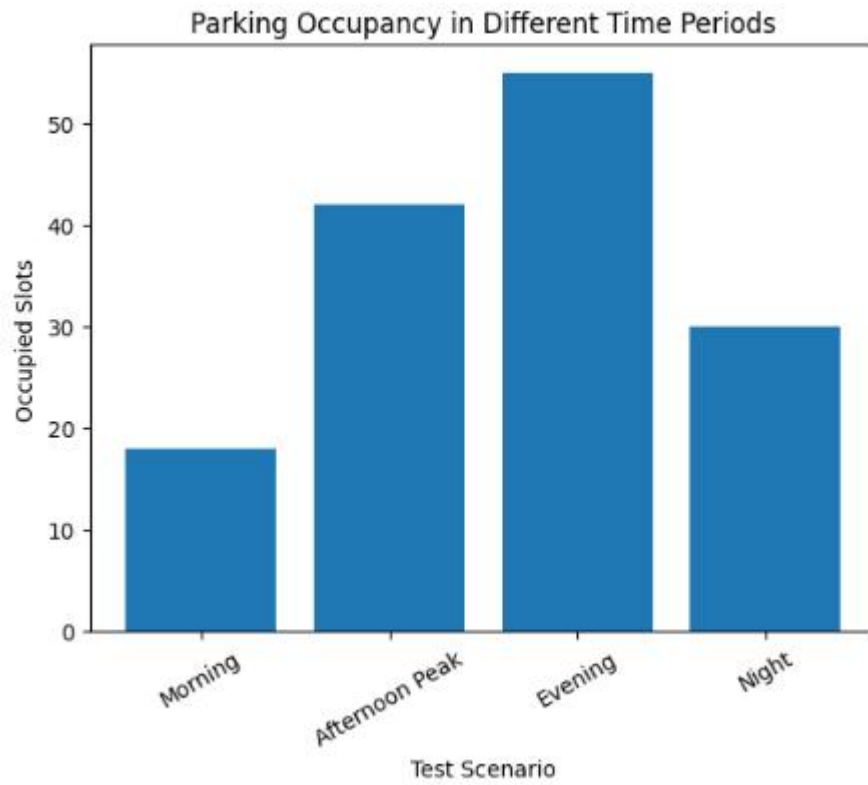


Figure 7. 2 Parking Occupancy During Different Time Periods

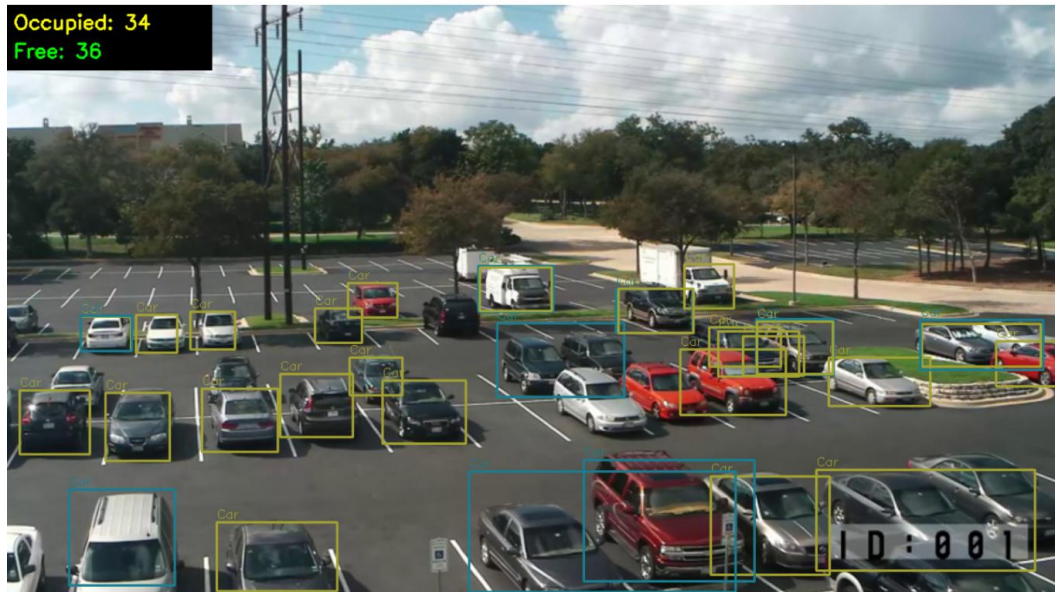


Figure 7. 3 Parking lot visuals during morning hours



Figure 7. 4 Parking lot visuals during afternoon hours

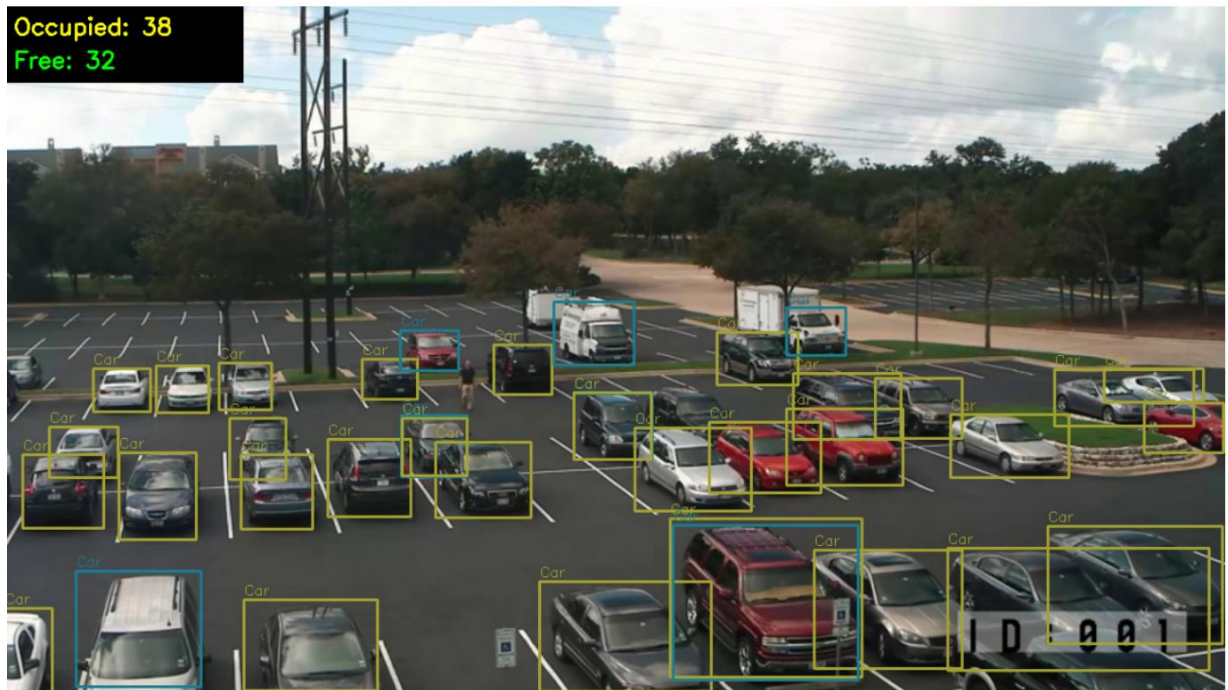


Figure 7.5 Parking lot visuals during evening hours

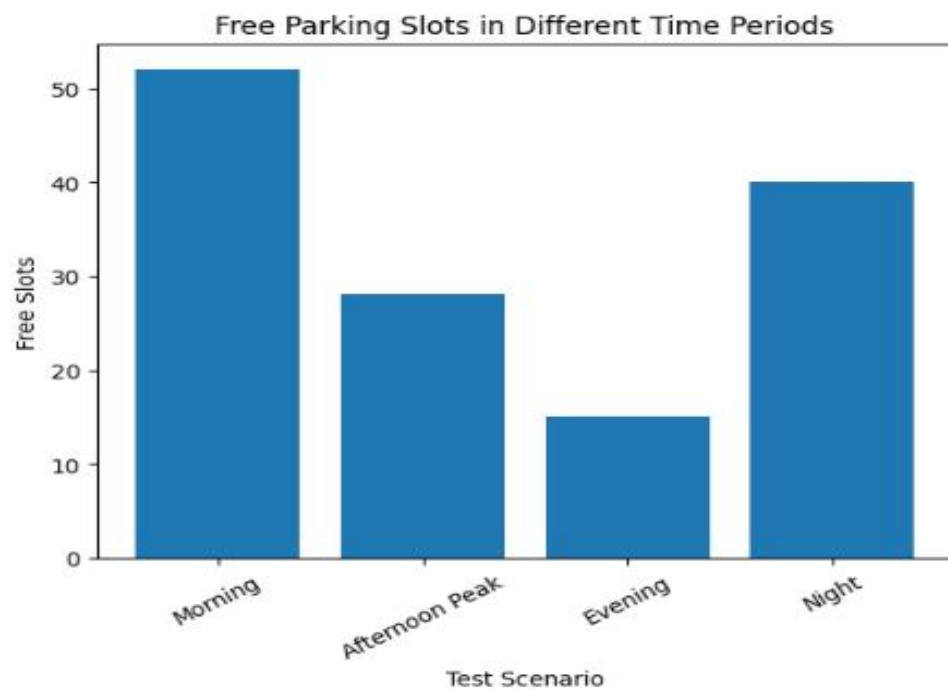


Figure 7.6 Availability of Free Slots Across Test Scenarios

Table 7. 3 System Performance Metrics

Parameter	Observed Value
Average Detection Accuracy	94.2%
Average Processing Time per Frame	0.7 seconds
Average System Response Time	1.0 second
Detection Confidence Threshold	0.15
Total Parking Capacity	70 Slots

The identified performance was also illustrated based on bounding boxes on the video frames of vehicles that gave a characteristic view of occupied parking spaces.

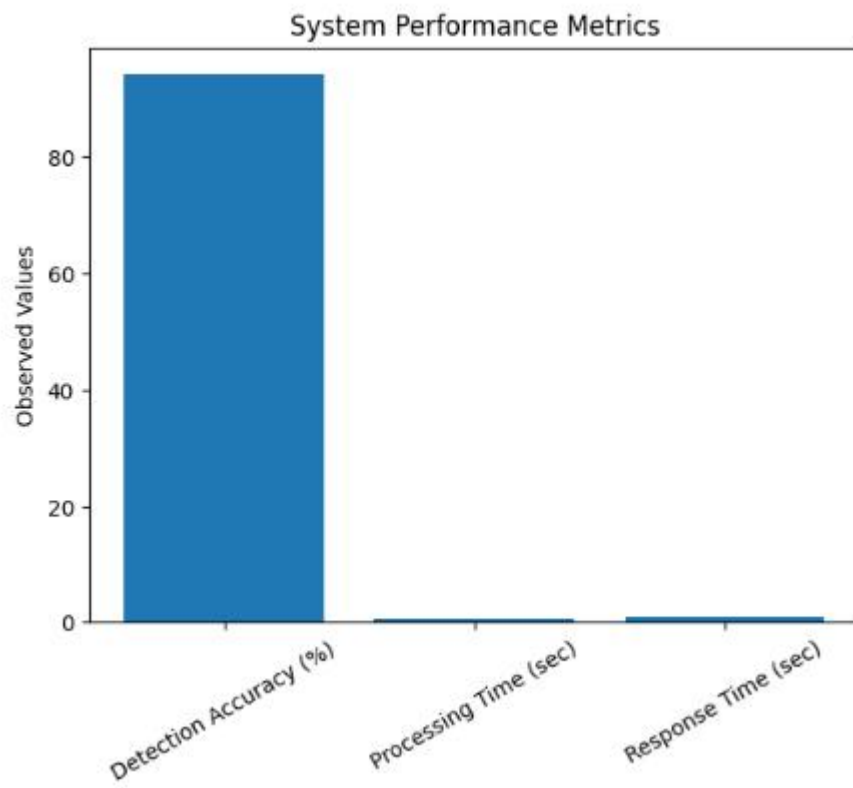


Figure 7. 7 Overall System Performance Evaluation

7.2 Performance Evaluation

The functionality of the proposed system was tested against a number of parameters based on the results of real-time detection.

Detection Accuracy : The model of Mask R-CNN demonstrated stable results in the detection of vehicles in a parking lot. The model employed pixel-level segmentation to identify the cars in all situations including those when several vehicles were moving side by side.

Efficiency in Real Time Processing : Multi-threading was used to divide video playback and detection processes in the system. This enabled videos streaming and the frames to be processed by the detection model in the background. This design has been very helpful in terms of responsiveness of the system and smooth visualization.

Occupancy Estimation: the formula adopted was to estimate the parking occupancy:

Occupied Slots = Slots occupied by vehicles can be Counted.

Free Slots = Parking Capacity-Occupied Slots

The system was live and aligned the parking position and displayed the outcome on firebase through which real time parking data was viewable

7.2.1 System Stability

The system had the capability to give a regular output on diverse video resolutions and light settings. It was also good in retrieving vehicles even during small occlusions and even changes in shadows.

7.2.2 Performance of Cloud Integration

The Firebase connection supported real-time shadowing of the parking information and it was thus feasible to get the data on the parking availability remotely.

7.3 Key Findings

Implementation and evaluation of the proposed efficient and clever Street Parking Management System led to certain significant conclusions. In the testing process, it developed to be determined that the Mask R-CNN model was useful in identifying vehicles based on the use of segmentation method was based on instance segmentation was effective even in the cases when several vehicles emerged.

in the same frame. The model proved very adaptable when used in varied environmental conditions which helped to get reliable occupancy prediction.

A threaded system architecture contributed to the enhancement of the efficiency of operations since, with the system, video streaming and the detection processes can be performed simultaneously. The parallel implementation guaranteed easier performance and less processing delays hence supporting near real time monitoring. Dynamically calculated parking occupancy rate was calculated by the number of vehicles detected and this gave a viable and automated method of calculating slot availability.

Consequently, one would be able to remotely monitor the availability of parking without the need to manually update the availability and refresh a page. The other benefit of the system was its ability to identify a variety of vehicle types, such as cars, motorcycles, buses, and trucks, which increases its relevance in a wide range of city environments.

Also, the automated system of detection and monitoring decreased the necessity of human management of the processes greatly and enhanced the efficiency of the operations and reduced the risk of human error. The system is web based and may be scaled in terms of size because it was developed using scalable technologies, hence it can be expanded to fit bigger parking systems in the smart city settings. On the whole, the results have shown that the solution suggested is effective, flexible and can be implemented in the current systems of urban traffic management.

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The Smart Parking Management System created within the scope of the given project illustrates that deep learning-based methods are effectively used in order to detect and monitor the parking space in real-time. The system employs the Mask R-CNN (Mask Region-Based Convolutional Neural Network) framework to segment the parking lot video feeds by instances and thus is able to detect and classify vehicles within the predefined parking slots. In comparison to classic sensor-based or IoT-based parking systems, this type of approach is fully based on computer vision, therefore, minimizing the hardware and the cost of infrastructure.

Mask R-CNN was found to be very adequately applicable in this application because it not only detects but also provides a segmentation mask that is accurate to each vehicle detected. This enabled the system to precisely identify the occupancy or vacancy of a parking slot by comparing the found vehicle masks with the designated parking zones. The instance segmentation technique also promoted the accuracy of the system relative to the traditional object detection patterns, which use bounding boxes.

The system received training and testing on the video data in parking lots under different conditions of light and the environment. The experiment results demonstrated the good detecting results, the consistent performance in the real-time processing, and the sound recognition of the free and occupied slots. The model was resistant to small occlusions as well as different orientations of vehicles. The findings affirm that the computer vision models of deep learning can be used successfully to substitute the conventional manual surveillance and sensor-specific approaches.

Besides, the created system enhances the productivity of managing parking as it includes real-time availability data. It has the benefit of minimizing unwanted movements of vehicles in parking stations, congestion in the traffic, as well as a part in reducing the amount of fuel burnt

and carbon emissions. Scalability The system architecture can be customized to the requirements of other parking environments, including shopping malls, offices, universities and government or community parking facilities.

To sum up, smart parking management based on Mask R-CNN can be successfully used to create a balance between the old-fashioned and the new intelligent automated parking systems. The project confirms that the vision systems based on deep learning offer an alternative that is practical, cost-effective, and scalable to the current urban parking management.

8.2 Future Scope

Despite the fact that the existing system has attained good vehicle recognition and parking slots location, it is still possible to make a lot of improvements and apply it to real-life applications. The further step of the development of this system can be adjusted to smarter options to develop capability and achieve efficiency, usability, and scalability.

One of the potential possibilities can be the introduction of AI-based models of parking prediction improvements. The machine learning algorithms can be deployed to generate peak hours and the availability of slots in future based on the historical parking occupation records. This would allow users to plan their visits in a more efficient and efficient way also enable the parking administrators to use the available space is used in a better manner. A predictive analytics can also transform the system to be a more proactive than a reactive monitoring tool solution of parking management.

The other possible extension is the use of the system in combination with the payment gateway and reservation system. It allowed the user to book parking slots earlier using a mobile app, pay online and get real-time information on the availability of slots. This would result in a complete automation of smart parking system that minimizes human intervention and also improves the ease of use.

GPS navigation support can also be introduced to improve the system. Parking management system can be integrated with the navigation services whereby the drivers can be directed to

the closest free parking slot. This would greatly decrease internal traffic in the parking premises and enhance increased operational efficiency.

On the technical enhancements, future studies can aim at training the Mask R-CNN model in fast inference by utilizing lightweight architectures or edge computing devices. Implementing the model on embedded systems or cloud tools would allow implementing some of the parking sites at large scale. Also, it is possible to combine the multi-camera systems to enhance the detection rate within the big or complicated parking area.

The future developments can involve the implementation of license plate recognition to enable automatic logging of vehicles and improvement of their security. The use of an instance segmentation algorithm with vehicle tracking algorithms might also aid in tracking the movement pattern of vehicles and in avoiding illegal parking.

Overall, the future of Smart Parking System is extensive. Since the sphere of deep learning, computer vision, and intelligent transportation systems continues to advance, the specified model can be evolved into a comprehensive solution of intelligent mobility that will contribute to sustainable urbanization and an efficient approach to traffic control.

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Research Paper Acceptance Proof



Fw: Acceptance with Minor Revision: Please Register and Resubmit your Revised paper(Paper Id: 238)

1 message

Akash Goel <a.goel54@yahoo.com>

Thu, 30 Oct, 2025 at 5:13 pm

Reply to: Akash Goel <a.goel54@yahoo.com>

To: anusha.2226cs1100@kiet.edu <anusha.2226cs1100@kiet.edu>

Yahoo Mail: Search, organise, conquer

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From: "Microsoft CMT" <noreply@msr-cmt.org>

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Cc:

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Dear Akash Goel,

Thanks for submitting your paper for 2026 International Conference on Computing Sciences and Communications (ICCS) organized by ABES Engineering College, Ghaziabad, India.

All the submitted papers go through a rigorous review process after a plagiarism check and we have received the comments from the Technical Program Committee (TPC) regarding your submission "Smart Street Parking Management Using Real-Time Computer Vision and YOLOv8" (Paper ID: 238).

Reviewer's comments:

Reviewer 1:

1. The abstract is too lengthy and should be rewritten in a concise and appropriate scientific manner, clearly highlighting the key objectives, methods, and contributions.
2. The paper contains excessive theoretical explanation, but lacks real-time images or visual evidence of the installed architecture, which are important to validate the practical implementation.
3. The literature referred to in the paper is outdated. It is strongly recommended that the author review recent papers published between 2022 and 2025 to identify current research gaps.
4. The research gaps and author contributions are not clearly presented. The author should analyze recent works and present a comparative analysis to justify the novelty and relevance of their approach.

Reviewer 2:

1. Despite the robust technical contributions, the paper can be made stronger with increased detail about data privacy, video anonymization schemes, and regulative compliance (i.e., GDPR) to achieve real-world scalability.
2. Further explanation of ablation analysis between the pruning, quantization, and knowledge distillation effect on accuracy as well as latency would be beneficial to the empirical clarity.
3. The figures could be made slightly better in terms of consistency as well as readability.

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Proof of patent publication

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(57) Abstract : The present invention relates to a smart and effective management system for street parking that enables real-time parking space discovery and reservation without using physical sensors. It utilizes a web-based platform integrated with real-time data analytics, predictive modeling, and dynamic pricing mechanisms. This system aims to reduce parking-related congestion, enhance urban mobility, and optimize street parking utilization. It utilizes machine learning typically works by using cameras to capture images of a parking lot, then applying computer vision algorithms to identify occupied and vacant parking spaces in real-time, with this data relayed to users through a mobile app or web interface, allowing them to easily find available parking spots.		
No. of Pages : 12 No. of Claims : 2		